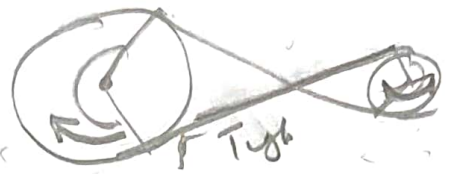
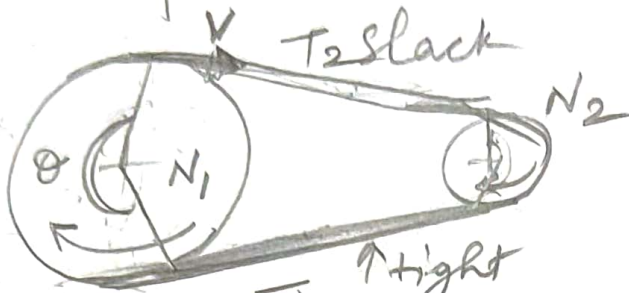


Open Belt

Crossed belt



driver

$$\frac{T_1}{T_2} = e^{\mu \theta}$$

↑ tight
radian

Ratio of belt tension

$$V = \frac{\pi D N}{60} \text{ m/s}$$

$$V_1 = \frac{\pi D_1 N_1}{60}$$

$$V_2 = \frac{\pi D_2 N_2}{60}$$

$$V_1 = V_2$$

No slip

$$D_1 N_1 = D_2 N_2$$

$$\frac{N_1}{N_2} = \frac{D_2}{D_1}$$

Length of open belt



$$\theta_1 = \pi + 2\beta$$

$$\cos \beta = \frac{AB}{x}$$

$$\theta_2 = \pi - 2\beta$$

$$L = r_1 \theta_1 + AB + r_2 \theta_2 + CD$$

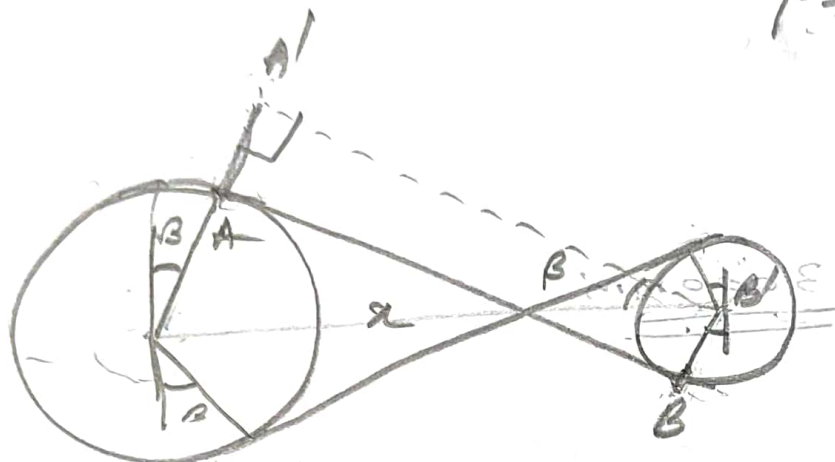
$$= r_1 (\pi + 2\beta) + x \cos \beta + r_2 (\pi - 2\beta) + x \cos \beta$$

$$L_0 = \pi(r_1 + r_2) + 2\beta(r_1 - r_2) + 2x \cos \beta$$

$$\sin \beta = \frac{r_1 - r_2}{x}$$

rad.

$$\cos \beta = \sqrt{1 - \sin^2 \beta}$$



$$\theta_1 = \pi + 2\beta$$

$$L_0 = \pi(r_1 + r_2) + 2\beta(r_1 - r_2) + 2x \left[1 - \frac{1}{2} \left(\frac{r_1 - r_2}{x} \right)^2 \right]$$

$$L_0 = \pi(r_1 + r_2) + 2\beta(r_1 - r_2) + 2x - \frac{(r_1 - r_2)^2}{x}$$

$$L_c = r_1(\pi + 2\beta) + r_2(\pi + 2\beta) + 2 A'B'$$

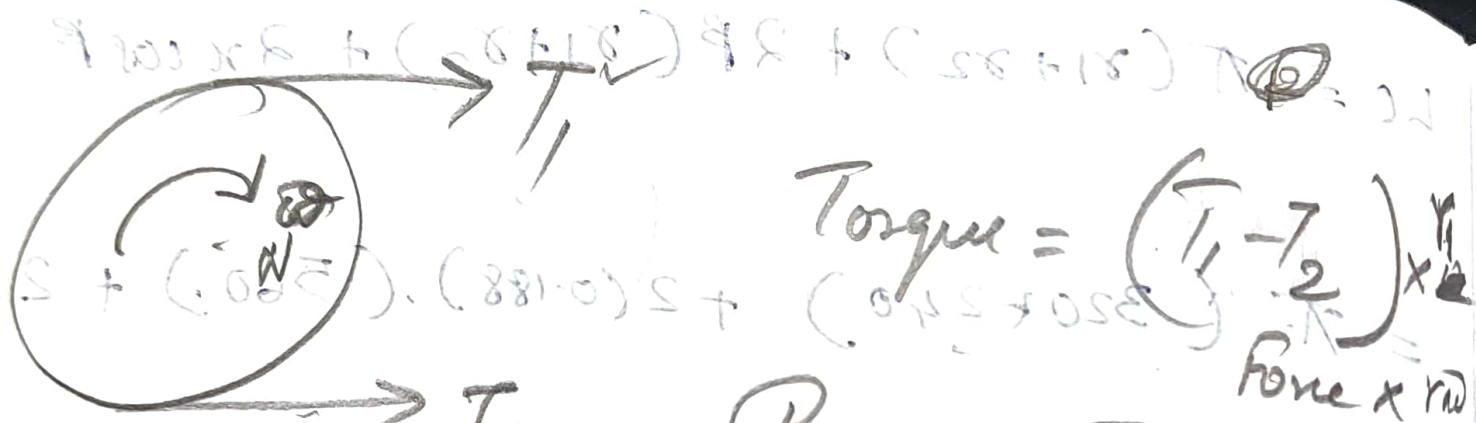
$$= \pi(r_1 + r_2) + 2\beta(r_1 + r_2) + 2x \cos \beta$$

$$\cos \beta = \frac{A'B'}{x}$$

$$A'B' = x \cos \beta$$

$$\sin \beta = \frac{r_1 - r_2}{x}$$

~~Power~~ Torque =



$$\text{Torque} = (T_1 - T_2) \times r$$

Force $\times$ rad

$$\text{Power} = \text{Torque} \times \omega$$

$$= (T_1 - T_2) \times r \times \frac{2\pi N}{60}$$

$$= (T_1 - T_2) \times \frac{\pi N}{60}$$

$$L = \pi(r_1 + r_2) + 2p(r_1 + r_2) + 2\pi r \cos \beta$$

$$\sin \beta = \frac{r_1 + r_2}{r}$$

$$= \frac{325}{1950} = 0.1667$$

$$\beta = \sin^{-1}(0.1667)$$

$$= 9.59^\circ$$

$$= 9.5 \times \frac{\pi}{180} \text{ (rad)}$$

Slip

5%

$$\frac{\pi DN}{60}$$

$$\left(\frac{S}{100}\right) \times \frac{\pi DN}{60}$$

$$= \frac{\pi DN}{60} \left[1 - \frac{S}{100}\right]$$

$$\text{circ 81 PA} =$$

$$R = (\text{tooth of contact}) \times 9.5 \times \pi =$$